

f_g is not uniformly continuous

exercise 14; 15: dans VT.

exercise 16, 17: examen tas am li jah

Sheet n° 06:

exo c1:

• f is derivable on $] -1, 1[$, because the composition, and product of derivable function $(\exp(x), x^2)$.

and $\forall x \in (-1, 1)$, $f'(x) = 2x e^{-x^2} (1 - x^2)$

on $] -\infty, -1[\cup] 1, +\infty[$, the derivative of f equal to 0.

$$\text{at } x_0 = 1$$
$$\lim_{x \rightarrow 1} \left(\frac{f(x) - f(1)}{x - 1} \right) = \lim_{x \rightarrow 1} \left(\frac{x^2 e^{-x^2} - e^{-1}}{x - 1} \right) = 0$$

$$\lim_{x \rightarrow 1} \left(\frac{f(x) - f(1)}{x - 1} \right) = \lim_{x \rightarrow 1} \frac{\frac{1}{e} - \frac{1}{e}}{x - 1} = 0.$$

so f is derivable at $x = 1$.

since f is derivable $1 + x = 2$ and $f(-1) = 0$

consequently, f is derivable on $\mathbb{R}$, and:

$$f'(x) = \begin{cases} 2xe^{-x^2}(1-x^2); & |x| < 1. \\ 0 & ; |x| \geq 1. \end{cases}$$

ex 002:

23)

$$\lim_{x \rightarrow 0} \left(\frac{g(x) - g(0)}{x - 0} \right) = \lim_{x \rightarrow 0} \left(\frac{x^4 - 0}{x - 0} \right) = 0$$

Hence, derivable at $x_0 = 0$ and $g'(0) = 0$
the same way for $g^{(2)}(0)$ and $g^{(3)}(0)$.

ex 003:

$$1. \quad f'(x) = 2e^{2x} \sin(x) + e^{2x} \cos(x).$$

$$2. \quad f''(x) = 2f'(x) + 2e^{2x} \cos(x) - e^{2x} \sin(x)$$

$$3. \quad \text{Leibniz Formula, } (fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x)$$

$$= \binom{n}{0} f^{(0)}(x) g^{(n)}(x) + \dots + \binom{n}{n} f^{(n)}(x) g^{(0)}(x).$$

$$\text{we know that } \sin^{(n)}(x) = \sin\left(x + n \frac{\pi}{2}\right)$$

$$\text{we have: } \sin^{(n)}(x) = \cos(x) = \sin\left(x + \frac{\pi}{2}\right)$$

$$\text{and: } \sin^{(2)}(x) = -\sin(x) = \sin(x + \pi) = \sin\left(x + 2 \frac{\pi}{2}\right)$$

$$\sin^{(n)}(x) = \sin\left(x + n \frac{\pi}{2}\right).$$

concerning e^{2x} , we have the n th derivative

equal to $2^n e^{2x}$.

Applying Leibniz formula, we get:

$$(e^{2x} \sin(x))^{(n)} = 2^n e^{2x} \sin\left(x + n \frac{\pi}{2}\right) + \dots + 1 e^{2x} 2^n \sin(x).$$

$$e^{2x} \sum_{k=0}^n \binom{n}{k} 2^k \sin\left(x + (n-k) \frac{\pi}{2}\right).$$

exo 04:

$$f(x) = x^3 - 3x^2 + 4x \quad D_f = [0, 2].$$

a. Since f is a Polynomial function, so it's continuous on D_f and differentiable on $(0, 2)$, so, MVT can be applied.

$$b. f(2) = 4, f(0) = 0$$

$$f'(x) = 3x^2 - 6x + 4.$$

Let $c \in (0, 2)$ and consider:

$$f'(c) = \frac{f(2) - f(0)}{2 - 0}$$

$$3c^2 - 6c + 4 = 2.$$

$$3c^2 - 6c + 2 = 0.$$

$$\Delta = 36 - (4 \times 6) = 36 - 24 = 12.$$

$$c_1, c_2 = \frac{6 \pm \sqrt{12}}{6} = 1 \pm \frac{\sqrt{3}}{3}.$$

exd 05:

$$a) \frac{d}{dx} (x^2 + y^2 = 25)$$

$$\Rightarrow 2x + 2y'y = 0$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

b) the slope of the tangent line at the point $(3, 4)$ is: $-\frac{3}{4}$.

$$5/ f(x) = \ln(x)$$

2) equation of the tangent to

$$y = f'(x)(x-2) + f(2)$$

$$y = \frac{1}{x}x - \frac{2}{2} + \ln(2)$$

is the normal line so we

$$\vec{v} \cdot \vec{u} = 0 \quad \vec{v} \begin{pmatrix} 1 \\ a \end{pmatrix} \quad \vec{u} \begin{pmatrix} 1 \\ a \end{pmatrix}$$

$$so \quad 1 + 0 \cdot a = 0 \quad a = -\frac{1}{0}$$

$$y = -2x + b$$

$$-2(2) + b = \frac{1}{2}(2) = 1 + \ln$$

$$b = \ln 2 + 4$$

$$y = -2x + (\ln 2 + 4)$$

ex 07:

a, Note that f is differentiable at $x=0$ ^{only}

If $\{a_n\}$ is rational sequence different to 0 and converging to 0.

$$\lim_{n \rightarrow \infty} \frac{f(a_n) - f(0)}{a_n - 0} = \lim_{n \rightarrow \infty} \frac{a_n^2}{a_n} = \lim_{n \rightarrow \infty} a_n = 0.$$

If $\{r_n\}$ is irrational sequence converging to 0 then:

$$\lim_{n \rightarrow \infty} \left(\frac{f(r_n) - f(0)}{r_n - 0} \right) = \lim_{n \rightarrow \infty} \left(\frac{0 - 0}{r_n - 0} \right) = 0$$

Thus f is differentiable at 0, and $f'(0) = 0$

ex 08:

$$\lim_{k \rightarrow \infty} \left[\frac{f(x) - f(0)}{x - 0} + \frac{1}{2} \frac{f(\frac{x}{2}) - 0}{\frac{x}{2} - 0} + \dots + \frac{f(\frac{x}{k}) - f(0)}{\frac{x}{k} - 0} \right]$$

we put, $\forall k \in \mathbb{N}$, $\frac{x}{k} = y$ when $x \rightarrow 0$
 $y \rightarrow 0$

$$\text{so } \lim_{y \rightarrow 0} \frac{f(y) - 0}{y - 0} = f'(0), \text{ because of differentiability}$$

at 0.

thus:

$$\lim_{k \rightarrow \infty} \left[\frac{1}{k} \left(f(x) + f\left(\frac{x}{2}\right) + \dots + f\left(\frac{x}{k}\right) \right) \right] = f'(0) \sum_{i=1}^k \frac{1}{i}$$

Ex 9

$$f(c) = 0$$

f is diff at $c \iff$

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

$$h = \frac{1}{n}$$

$h \rightarrow 0$ then $n \rightarrow \infty$

$$f'(c) = \lim_{n \rightarrow \infty} \frac{f(c + \frac{1}{n}) - f(c)}{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left(f\left(\frac{x}{n}\right) + f\left(\frac{x}{n}\right) + \dots + f\left(\frac{x}{n}\right) \right)$$

put $h = \frac{x}{n}$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{x} \sum_{h=1}^n f\left(\frac{x}{h}\right) \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{h=1}^n \frac{f\left(\frac{x}{h}\right)}{h \cdot \frac{x}{h}}$$

$$= \lim_{n \rightarrow \infty} \sum_{h=1}^n \frac{1}{h}$$

$$= \sum_{h=1}^n \frac{1}{h} \lim_{n \rightarrow \infty}$$

$$= \sum_{h=1}^n \frac{1}{h} \lim_{h \rightarrow 0} \frac{f}{h}$$

$$\exists c \in]x, y[;$$

$$f(c) = f(x) - f(y)$$

$$- \sin c = \frac{\cos x - \cos y}{x - y}$$

$$\sin c = \frac{|\cos x - \cos y|}{|x - y|} \Rightarrow$$

$$\forall x \in \mathbb{R}$$

$$|\Delta f_n| = \|f_n\|$$

$$\frac{|\cos x - \cos y|}{|x - y|} \leq 1$$

$$\Rightarrow \frac{|\cos x - \cos y|}{|x - y|} \leq 1$$

exo 10:

Since f' exists and is continuous on $[a, b]$, then f' is bounded $\Leftrightarrow \exists K \in \mathbb{R}, \forall x \in [a, b]$

$$|f'(x)| \leq K$$

Let $x, y \in [a, b]$, f is continuous $[a, b]$ and differentiable on $]a, b[$. From the MVT

$$\exists c \in]x, y[: \frac{|f(x) - f(y)|}{|x - y|} = |f'(c)| \leq K$$

$$\Rightarrow |f(x) - f(y)| \leq K|x - y|$$

Thus f is K -Lipschitz on $[a, b]$.

exo 11:

$$\text{let consider } f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x^2}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

f is uniformly continuous in $[0, 1]$ and derivable in $]0, 1[$.

$$f'(x) = 2x \sin\left(\frac{1}{x^2}\right) - \frac{2}{x} \cos\left(\frac{1}{x^2}\right)$$

Thus f' isn't bounded on $]0, 1[$.

exo 12: $f(x) = [x] \in \mathbb{R} - \mathbb{Z}$

$f'(x) = 0$ but f is not constant

let: $f: \mathbb{R}^* \rightarrow \mathbb{R}$

$$x \mapsto \begin{cases} 1; & x > 0 \\ -1; & x < 0 \end{cases}$$

f is continuous on A , and differentiable on A , and $f'(x) = 0, \forall x \in A = \mathbb{R}^*$ but f isn't constant.

exo 13:

$$f(x) = x^3 - 6x^2 + 9x + 2.$$

1. critical points:

$$f'(x) = 3x^2 - 12x + 9.$$

$$f'(x) = 0 \Rightarrow x^2 - 4x + 3 = 0.$$

$$\Delta = 4^2 - 4 \times 3 = 4.$$

$$x_1 = 1, \quad x_2 = 3.$$

the critical points are: 1, 3

then: $f(1)$ is local maximum and $f(3)$ is local minimum ($\forall x \in J, \forall x \in J$, we have $f(x) \leq f(1)$)

exo 14:

a. $f(x) = x^4 - 4x^2 + 3.$

we have f is continuous and differentiable on $(-2, 2)$ then:

$$f'(x) = 4x^3 - 8x.$$

the critical ~~of~~ points of f are the
solution of the equation $f'(x) = 0$

$$4x^3 - 8x = 0 \Rightarrow 4x(x^2 - 2) = 0$$

$$\text{So } \begin{cases} 4x = 0 \\ \text{or} \\ x^2 - 2 = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ x = -\sqrt{2} \text{ or } x = \sqrt{2} \end{cases}$$

the critical points are: $0, \sqrt{2}, -\sqrt{2}$.

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3 is global maximum on $[-2, 2]$

-1 is a global minimum on $[-2, 2]$.

exo 15:

$$(a) \quad f'(x) = -2x e^{-x^2}$$

the critical point is

$$x = 0$$

$$(b) \quad f''(x) = -2x e^{-x^2} - (2x e^{-x^2}) (-2x) \\ = -2e^{-x^2} + 4x^2 e^{-x^2}$$

$$f''(0) = -2 < 0$$

so $f(0)$ the maximum of f .

exercice 16 :

$$\text{we put : } \begin{cases} x = \text{length} \\ y = \text{width} \end{cases}$$

$$a. \quad 2x + y = 60 \iff y = 60 - 2x.$$

we have :

$$\begin{aligned} A &= x \cdot y \\ &= x \cdot (60 - 2x) \\ &= 60x - 2x^2 \end{aligned}$$

$$f(x) = 60x - 2x^2$$

b. $f(x)$ is continuous and differentiable,

$$f'(x) = -4x + 60.$$

$$\text{we solve } f'(x) = 0 \implies$$

$$f'(x) = 0 \iff -4x + 60 = 0$$

$$x = 15 \text{ (15 is critical point)}.$$

ii) - we have :

$$2 \cdot 15 + y = 60 \implies y = 30$$

Then The maximum area is

$$S = 15 \cdot 30.$$

exercice 17 :

$$a. \quad f'(x) = -\cos(x) + \sin(x)$$

$$f'(x) = 0 \iff \sin x = \cos x.$$

$$\text{so: } x = \frac{\pi}{4} \wedge x = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$b. \quad f\left(\frac{\pi}{4}\right) = \sqrt{2} \text{ (max)}, \quad f(2\pi) = 1.$$

$$f\left(\frac{5\pi}{4}\right) = -\sqrt{2} \text{ (min)}$$

Exercise 18:

$$a. \quad xy \leq \frac{x^p}{p} + \frac{y^q}{q}$$

meT 01

Let $f(x) = e^x$, $x \in \mathbb{R}$.

f is a convex function means that:

$\forall x_1, x_2 \in \mathbb{R}$ we have:

$$\lambda e^{x_1} + (1-\lambda)e^{x_2} \geq e^{\lambda x_1 + (1-\lambda)x_2}$$

$$\forall \lambda \in [0, 1]$$

If we take $\lambda = \frac{1}{p}$

$$\text{we get: } \frac{1}{p} e^{x_1} + \frac{1}{q} e^{x_2} \geq e^{\frac{1}{p} x_1 + \frac{1}{q} x_2}$$

We have $x > 0 \wedge y > 0$

so if we put $x_1 = p \ln x \wedge x_2 = q \ln y$

we get:

$$e^{\frac{1}{p} p \ln x + \frac{1}{q} q \ln y} \leq \frac{1}{p} e^{p \ln x} + \frac{1}{q} e^{q \ln y}$$

This give us

$$e^{\ln x + \ln y} \leq \frac{1}{p} x^p + \frac{1}{q} y^q$$

$$\Leftrightarrow xy \leq \frac{x^p}{p} + \frac{y^q}{q}$$

★ ex 18:

$$b. (x+y) \ln \left(\frac{x+y}{2} \right) \leq x \ln x + y \ln y.$$

$$g(x) = x \ln x$$

$$g''(x) = \frac{1}{x} > 0, \text{ so } g \text{ is conv.}$$

One has:

$$g\left(\frac{x+y+z+t+r+s}{6}\right) \leq \frac{1}{6}g(x) + \frac{1}{6}g(y) + \frac{1}{6}g(z) + \frac{1}{6}g(t) + \frac{1}{6}g(r) + \frac{1}{6}g(s)$$

$$\Rightarrow \frac{x+y+z+t+r+s}{6} \ln \left(\frac{x+y+z+t+r+s}{6} \right) \leq \frac{1}{6} (x \ln x + y \ln y + z \ln z + t \ln t + r \ln r + s \ln s)$$

$$\text{we Take, } x = z = r \\ y = t = s.$$

⋮

EXERCISE 19

a. $f(x) = \ln x$ concave :

$$\ln \left(\frac{\sum_{i=1}^n x_i}{n} \right) \geq \sum_{i=1}^n \frac{1}{n} \ln(x_i)$$

$$\geq \frac{1}{n} \sum_{i=1}^n \ln(x_i)$$

$$\geq \frac{1}{n} \ln \prod_{i=1}^n x_i \geq \ln \left(\prod_{i=1}^n x_i \right)^{1/n}$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n x_i \geq \sqrt[n]{\prod_{i=1}^n x_i}$$

$$\Rightarrow AM \geq GM$$

★ H-H inequality:

If f is convex on $[a, b]$ Then:

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq \frac{f(a)+f(b)}{2}$$

$$\frac{1}{b-a} \int_a^b e^x dx \leq \frac{e^a + e^b}{2}$$

$$\frac{e^b - e^a}{b-a} \leq \frac{e^a + e^b}{2}$$

$$e^b - e^a \leq (b-a) \left(\frac{e^b + e^a}{2} \right)$$

